Joseph R. Farah

Contact Information

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Citations: Google Scholar profile

Research Interests

I lead programs to measure the spins of compact objects across regimes, from infant millisecond magnetars in superluminous supernovae (SNe) to supermassive black holes resolved at the horizon scale. These measurements probe strong-field physics where no laboratory can reach and expose the engines driving the most energetic phenomena in the Universe.

Tags:

Astrophysical Phenomena: core-collapse SNe, superluminous SNe, black hole spacetimes, chirping SNe, photon rings
Observational Techniques: very-long-baseline interferometry (VLBI), radio astronomy, optical photometry, optical spectroscopy
Computational & Theoretical Methods: general relativity, Bayesian modeling, machine/deep learning, geometric modeling, interferometric imaging, shock-cooling models, light curve modeling

Education

University of California Santa Barbara, Santa Barbara, CA, USA 2021–2026
Ph.D. in Physics (completed) 2026
M.A. in Physics (completed) 2023

University of Massachusetts Boston, Boston, MA, USA 2017–2021
B.S. in Physics, summa cum laude, with distinction

Selected Honors and Awards

Miller Fellowship, Miller Institute for Basic Science, Berkeley 2026–2029
Award amount: $354,000

Bhaumik Prize Postdoctoral Fellowship, UCLA (declined) 2026
Award amount: $276,000

UC Santa Barbara Mananya Tantiwiwat Award 2025

Broida-Hirschfelder Fellowship 2025
citation: *In recognition of his project uncovering the secrets of the strangest object in the universe, the black hole photon ring.*

1st Place Award, UC Santa Barbara Grad Slam 2025
Round 1 (Physics and Chemistry)

UC Santa Barbara GSA Excellence in Teaching Award (nominated) 2025

UC Santa Barbara Academic Senate Outstanding Teaching Assistant Award (nominated) 2024

Event Horizon Telescope Early Career Award 2022
citation: *For his leadership and contribution to the dynamic imaging of the interferometric data used to image Sgr A*.*

LeRoy Apker Award 2021
citation: *For the invention of the selective dynamical imaging method, with applications for studying rapidly-varying black holes.*

UC Santa Barbara James and Mary Jo Hartle Graduate Fellowship 2021

National Science Foundation Graduate Research Fellowship — 2021–2026
AWARD AMOUNT: $138,000

Arthur W. Martin III Scholarship — 2021

Breakthrough Prize in Fundamental Physics (co-recipient) — 2020

Northrop Grumman Scholarship — 2020

Barry M. Goldwater Scholarship — 2019

National Science Foundation Diamond Achievement Award‡ — 2019

Alton J. Brann Endowed Scholarship — 2019

Smithsonian Fellowship — 2018, 2019

Oracle Fellowship — 2018, 2019

AJAS/AAAS Lifetime Fellowship — 2018

Chancellor's Scholarship, University of Massachusetts Boston — 2017–2021

2nd Place Award, Massachusetts State Science and Engineering Fair — 2017

1st Place Award, Massachusetts State Science and Engineering Fair — 2016

SELECTED WORK AND RESEARCH EXPERIENCE

Miller Institute for Basic Research in Science — 2026 - Present
Miller Research Fellow, UC Berkeley — Berkeley, CA
Hosts: Daniel Kasen, Raffaella Margutti

Black Hole Initiative — 2026
Black Hole Explorer Fellow, Harvard University — Cambridge, MA
Supervisor: Michael Johnson

Las Cumbres Observatory — 2021 - 2026
NSF Graduate Research Fellowship — Goleta, CA
Supervisor: Andy Howell

Harvard-Smithsonian Center for Astrophysics — 2020 - 2026
Member of the Black Hole Explorer Collaboration — Cambridge, MA
Supervisor: Michael Johnson, Alex Lupsasca

Smithsonian Fellowship with the Event Horizon Telescope — Cambridge, MA — 2018 - 2021
Supervisor: Michael Johnson

Quantum Computing Research Group — 2019 - 2021
Research Intern with Dr. Alioscia Hamma — Boston, MA

Harvard Laboratory for Particle Physics and Cosmology — 2017 - 2018
Research Intern with Dr. Melissa Franklin — Cambridge, MA

QBism Research Group — 2017 - 2019
Oracle Fellow with Dr. Christopher Fuchs — Boston, MA

Harvard School for Engineering and Applied Sciences — Summer 2016
Summer Research Intern mentored by Dr. Chinwendu Enyioha — Cambridge, MA

Tufts Department of Astrophysics Summer 2015
Summer Research Intern mentored by Dr. Anna Sajina — Medford, MA

Teaching and Mentoring

Invited workshop creator and lecturer, *Presentation presentation: a seminar on seminars*, Las Cumbres Observatory, Goleta, CA. January 2026.

Death Star Seminar, founder and lecturer. Informal graduate seminar series on supernova theory, Las Cumbres Observatory, Goleta, CA. Designed curriculum and delivered lectures. 2026.

Teaching Assistant, PHYS 23 (Electrostatics and waves), UC Santa Barbara. Fall 2024.
UC Santa Barbara GSA Excellence in Teaching Award (nominated)
UC Santa Barbara Academic Senate Outstanding Teaching Assistant Award (nominated)

Undergraduate research mentoring

Eyouel Abate (UC Berkeley) 2026 - Present
Austin Mei (UC Berkeley) 2026 - Present
Courtney Duong (UC Santa Barbara) 2024 - Present
Project: Convolutional neural network for extraction of the $n = 1$ black hole photon ring
Award: *Gene and Susan Lucas Undergraduate Research Scholarship*
Award: *Edison STEM Research Scholarship*
Franklin Myhre (UC Santa Barbara) 2024 - Present
Project: Development of a `python` library for automated image-domain subring extraction
Sanjit Masanam (UC Santa Barbara) 2024 - 2026
Award: *2025 Goldwater Scholarship (nominated)*
Award: *2025 Ernest F. Hollings Scholarship*
Songgun Lee (UC Santa Barbara) 2024 - 2026
Co-founder at Masterminding
Tazzy Imbabi (UC Santa Barbara) 2024
Project: Exploration of selective dynamical imaging applications to BHEX

Service

Referee:
The Astrophysical Journal
Astronomy & Astrophysics

Internal referee, Event Horizon Telescope Collaboration.

Invited presenter, Las Cumbres Observatory Science Advisory Committee, Goleta, CA. Invited to present on the observatory's high-energy astrophysics science program. August 2025.

Local Organizing Committee, EHT Collaboration Meeting. 2020.

Local Organizing Committee, 2nd EHT Imaging Workshop, Black Hole Initiative, Harvard University, Cambridge, MA. July 2018.

Software Development

`eht-imaging`. Contributor. Repository.

The Supernova Exchange (SNEx). Contributor. Live site link.

Invited Talks and Colloquia

Precessing magnetar central engine drives a superluminous supernova. Galactic Center Group, UC Los Angeles, Los Angeles, CA. April 2026.

The first 4 years of SN 1993J revisited. UC Berkeley, Berkeley, CA. September 2025.

Lense-Thirring precessing magnetar engine drives a superluminous supernova. Multi-RAPTOR

colloquium, UC Berkeley, Berkeley, CA. September 2025.

Inferring black hole spin from images of the photon ring. VandyGRAF Initiative, Vanderbilt University, Nashville, TN. September 2024.

What Lies Within: Imaging the Galactic Center. April APS Meeting, New York, NY. Invited session speaker. April 2022.

The First Image of a Black Hole. Skyscrapers, Inc., Seagrave Memorial Observatory, North Scituate, RI. 2021.

Very Long Baseline Interferometry and the Event Horizon Telescope. Astronomy 121, University of Massachusetts Boston, Boston, MA. 2019 (2), 2020 (2).

Seminars and Contributed Talks

Precessing magnetar central engine drives a superluminous supernova. METAL group, University of Virginia, Charlottesville, VA. August 2026.

Spin inference of compact objects across regimes. Greene research group, Princeton University, Princeton, NJ. July 2026.

Lense-Thirring precessing magnetar engine drives a superluminous supernova. KIPAC, Stanford University, Stanford, CA. November 2025.

Machine- and deep-learning-driven angular momentum inference from Black Hole Explorer observations of the $n = 1$ photon subring. *Topics in Gravitational Physics*, APS Global Physics Summit. March 2025. Event URL.

Current and future impacts of survey capabilities on the stripped-envelope problem. Kavli Institute for Theoretical Physics, Santa Barbara, CA. February 2025.

Shock-cooling constraints via early-time observations of the Type IIb SN 2022hnt. Kavli Institute for Theoretical Physics, Santa Barbara, CA. February 2025.

Seeing the unseeable: the first image of a black hole. University of Massachusetts Boston, Boston, MA. May 2019.

Imaging a black hole with the Event Horizon Telescope. Harvard University, Cambridge, MA. April 2019.

Internal and Institutional Talks

Spin inference of compact objects across regimes. Black Hole Initiative, Cambridge, MA. August 2026.

Precessing magnetar central engine drives a superluminous supernova. Narayan group, Center for Astrophysics | Harvard & Smithsonian, Cambridge, MA. July 2026.

General relativity meets a supernova for the first time. Las Cumbres Observatory colloquium, Goleta, CA. January 2026. Event URL.

Presentation presentation: a seminar on seminars. Workshop, Las Cumbres Observatory, Goleta, CA. January 2026.

Selective dynamical imaging of interferometric data and the second image of a black hole. *Astro Tea*, University of California, Santa Barbara, CA. June 2022.

Outreach and Public Engagement

General relativity beats the heart of a dying star. Public lecture, Santa Barbara Museum of Natural History. March 2026. Event URL.

Extragalactic science with the Vera C. Rubin Observatory. Public lecture, Santa Barbara Museum

of Natural History. August 2025. Event URL.

The Black Hole Explorer and the edge of the universe. Public lecture, Santa Barbara Museum of Natural History. April 2025. Event URL.

The Black Hole Explorer. *Astronomy on Tap*, Santa Barbara, CA. December 2024. Event URL.

What does an astronomer do? Canalino Elementary School, Carpinteria, CA. October 2024.

The biggest telescopes! Aliso Elementary School, Carpinteria, CA. October 2023.

Imaging the black hole at our galaxy's center. Public lecture, Santa Barbara Museum of Natural History. May 2023. Event URL.

LCO's top 9 tips for taking your best black hole photo. *Astronomy on Tap*, Santa Barbara, CA. November 2022. Event URL.

AskScience AMA Series: Event Horizon Telescope results on the Galactic Center. *Reddit*. May 2022. Article URL.

LCO's top 9 tips for taking your best black hole photo. Honors and AP physics classes, Medford High School, Medford, MA. 2022.

Filmography

Galison, P. L. (Director). 2021. *Black Holes: The Edge of All We Know.* Sandbox Films. View on Netflix.

Galison, P. L. (Director). 2019. *Portrait of a Shadow.* Sandbox Films.

Publications (first-author)

Stats (all papers):
Citations: 25,000+
Ref'd Papers: 140+
h-index: 41
i10-index: 90

8. **J. R. Farah**, Lupsasca, A., Quataert, E., Johnson, M. D. 2026. *Interferometric inference of black hole spin from photon ring size and brightness.* ApJ, 1003, 63.

7. **J. R. Farah**, Prust, L. J., Howell, D. A., *et al.*. 2026. *Lense-Thirring precessing magnetar engine drives a superluminous supernova.* Nature, 651, 321-325. **NB:** selected for cover of the March 12, 2026 issue.

6. **J. R. Farah**, Howell, D. A., Hiramatsu, D., *et al.*. 2026. *When IIb Ceases To Be: Bridging the Gap Between IIb and Short-plateau Supernovae.* ApJ, 998, 321.

5. **J. R. Farah**, Prust, L. J., Terreran, G., *et al.*. 2025. *The First 4 Years of SN 1993J Revisited: Geometric m-ring Modeling of the Radio Shell with Closure Quantities Only.* Submitted to ApJ, positive referee report received. arXiv e-prints, arXiv:2509.20601.

4. **J. R. Farah**, Davelaar, J., Palumbo, D., Johnson, M., Delgado, J. 2025. *Machine and Deep Learning-driven Angular Momentum Inference from BHEX Observations of the $n = 1$ Photon Ring.* ApJ, 983, 185.

3. **J. R. Farah**, Howell, D. A., Terreran, G., *et al.*. 2025. *Shock-cooling Constraints via Early-time Observations of the Type IIb SN 2022hnt.* ApJ, 984, 60.

2. [S] **J. R. Farah**, Galison, P., Akiyama, K., *et al.*. 2022. *Selective Dynamical Imaging of Interferometric Data.* ApJL, 930, L18. arXiv:2409.08321.

1. **J. R. Farah**, Pesce, D. W., Johnson, M. D., Blackburn, L. 2020. *On the Approximation of the Black Hole Shadow with a Simple Polar Curve.* ApJ, 900, 77.

[S] This was the seventh paper of the 10-paper first Galactic Center results series from the Event Horizon Telescope, presenting the second image of a black hole and the first horizon-scale images of Sgr A*.

PUBLICATIONS (MAJOR OR NORMAL CONTRIBUTION)

(last updated: Sep 2025)

110. Wang, Z. Y., Pastorello, A., Cai, Y. Z., ..., **J. R. Farah**, *et al.*. 2025. *Massive stars exploding in a He-rich circumstellar medium: XI. Diverse evolution of five Ibn SNe 2020nxt, 2020taz, 2021bbv, 2023utc, and 2024aej.* A&A, 700, A156. arXiv:2506.15139.

109. Bostroem, K. A., Valenti, S., Sand, D. J., ..., **J. R. Farah**, *et al.*. 2025. *Late-time Hubble Space Telescope Ultraviolet Spectra of SN 2023ixf and SN 2024ggi Show Ongoing Interaction with Circumstellar Material.* arXiv e-prints, arXiv:2508.11756.

108. Kwok, L. A., Singh, M., Jha, S. W., ..., **J. R. Farah**, *et al.*. 2025. *JWST and Ground-based Observations of the Type Iax Supernovae SN 2024pxl and SN 2024vjm: Evidence for Weak Deflagration Explosions.* ApJL, 989, L33. arXiv:2505.02944.

107. Gagliano, A., Villar, V., Matsumoto, T., ..., **J. R. Farah**, *et al.*. 2025. *Evidence for an Instability-induced Binary Merger in the Double-peaked, Helium-rich Type IIn Supernova 2023zkd.* ApJ, 989, 182. arXiv:2502.19469.

106. Śniegowska, M., Trakhtenbrot, B., Makrygianni, L., ..., **J. R. Farah**, *et al.*. 2025. *AT 2019aalc: A Bowen Fluorescence Flare with a Precursor Flare in an Active Galactic Nucleus.* ApJ, 989, 173. arXiv:2505.00083.

105. Makrygianni, L., Arcavi, I., Newsome, M., ..., **J. R. Farah**, *et al.*. 2025. *The Double Tidal Disruption Event AT 2022dbl Implies that at Least Some "Standard" Optical Tidal Disruption Events Are Partial Disruptions.* ApJL, 987, L20. arXiv:2505.16867.

104. Dahale, R., Cho, I., Moriyama, K., ..., **J. R. Farah**, *et al.*. 2025. *Origin of the ring ellipticity in the black hole images of M87*.* A&A, 699, A279. arXiv:2505.10333.

103. Park, S. H., Rho, J., Yoon, S. C., ..., **J. R. Farah**, *et al.*. 2025. *Near-Infrared Spectroscopy and Detection of Carbon Monoxide in the Type II Supernova SN 2023ixf.* arXiv e-prints, arXiv:2507.11877.

102. Goddi, C., Carlos, D. F., Crew, G. B., ..., **J. R. Farah**, *et al.*. 2025. *First polarization study of the M87 jet and active galactic nuclei at submillimeter wavelengths with ALMA.* A&A, 699, A265. arXiv:2505.10181.

101. Jiang, S. Q., Xu, D., van Hoof, A. P., ..., **J. R. Farah**, *et al.*. 2025. *EP240801a/XRF 240801B: An X-Ray Flash Detected by the Einstein Probe and the Implications of Its Multiband Afterglow.* ApJL, 988, L34. arXiv:2503.04306.

100. Sun, H., Li, W. X., Liu, L. D., ..., **J. R. Farah**, *et al.*. 2025. *A fast X-ray transient from a weak relativistic jet associated with a type Ic-BL supernova.* Nature Astronomy, 9, 1073-1085. arXiv:2410.02315.

99. Kumar, H., Berger, E., Blanchard, P. K., ..., **J. R. Farah**, *et al.*. 2025. *A Near-infrared Search for Helium in the Superluminous Supernova SN 2024ahr.* ApJ, 987, 127. arXiv:2501.01485.

98. Singh, M., Kwok, L. A., Jha, S. W., ..., **J. R. Farah**, *et al.*. 2025. *Photometry and Spectroscopy of SN 2024pxl: A Luminosity Link Among Type Iax Supernovae.* arXiv e-prints, arXiv:2505.02943.

97. Jacobson-Galán, W., Dessart, L., Davis, K., *et al.*. 2025. *Final Moments III: Explosion Properties and Progenitor Constraints of CSM-Interacting Type II Supernovae.* arXiv e-prints, arXiv:2505.04698.

96. Liu, M., Zhang, Y., Wang, Y., ..., **J. R. Farah**, *et al.*. 2025. *Detection of an Orphan X-Ray Flare from a Blazar Candidate EP240709a with the Einstein Probe*. ApJ, 984, 5. arXiv:2412.18463.

95. Shrestha, M., DeSoto, S., Sand, D. J., ..., **J. R. Farah**, *et al.*. 2025. *Spectropolarimetry of SN 2023ixf Reveals Both Circumstellar Material and an Aspherical Helium Core*. ApJL, 982, L32. arXiv:2410.08199.

94. Ravi, A. P., Valenti, S., Dong, Y., ..., **J. R. Farah**, *et al.*. 2025. *Luminous Type II Short-plateau SN 2023ufx: Asymmetric Explosion of a Partially Stripped Massive Progenitor*. ApJ, 982, 12. arXiv:2411.02493.

93. Röder, J., Wielgus, M., Lobanov, A. P., ..., **J. R. Farah**, *et al.*. 2025. *A multifrequency study of sub-parsec jets with the Event Horizon Telescope*. A&A, 695, A233. arXiv:2501.05518.

92. Event Horizon Telescope Collaboration, ..., **J. R. Farah**, *et al.*. 2025. *The persistent shadow of the supermassive black hole of M87: II. Model comparisons and theoretical interpretations*. A&A, 693, A265.

91. Yesmin, N., Pellegrino, C., Modjaz, M., ..., **J. R. Farah**, *et al.*. 2025. *Spectral dataset of young type Ib supernovae and their time evolution*. A&A, 693, A307. arXiv:2409.04522.

90. Onori, F., Nicholl, M., Ramsden, P., ..., **J. R. Farah**, *et al.*. 2025. *The case of AT2022wtn: a tidal disruption event in an interacting galaxy*. MNRAS, 540, 498-520. arXiv:2504.21686.

89. Ma, X., Wang, X., Mo, J., ..., **J. R. Farah**, *et al.*. 2025. *Supernovae at distances ¡40 Mpc: II. Supernova rate in the local Universe*. A&A, 698, A306. arXiv:2504.04507.

88. Ma, X., Wang, X., Mo, J., ..., **J. R. Farah**, *et al.*. 2025. *Supernovae at distances ¡40 Mpc: I. Catalogues and fractions of supernovae in a complete sample*. A&A, 698, A305. arXiv:2504.04393.

87. Zeng, X., Luo, Y., Wang, X., ..., **J. R. Farah**, *et al.*. 2025. *SN 2023xqm: A gradually fading Ia supernova exhibiting isolated high-velocity signatures*. A&A, 698, A70.

86. Zeng, X., Hu, L., Zheng, S., ..., **J. R. Farah**, *et al.*. 2025. *SN 2023ehl: A Normal Type Ia Supernova with High-velocity Features*. ApJ, 986, 68.

85. Hsu, B., Smith, N., Goldberg, J. A., ..., **J. R. Farah**, *et al.*. 2025. *One Year of SN 2023ixf: Breaking through the Degenerate Parameter Space in Light-curve Models with Pulsating Progenitors*. ApJ, 990, 148. arXiv:2408.07874.

84. Palit, B., Śniegowska, M., Markowitz, A., ..., **J. R. Farah**, *et al.*. 2025. *Markarian 590: the AGN awakens*. MNRAS, 540, L14-L20. arXiv:2501.07225.

83. Subrayan, B. M., Sand, D. J., Bostroem, K. A., ..., **J. R. Farah**, *et al.*. 2025. *Early Shock Cooling Observations and Progenitor Constraints of Type IIb Supernova SN 2024uwq*. ApJL, 990, L68. arXiv:2505.02908.

82. Shu, X., Yang, L., Yang, H., ..., **J. R. Farah**, *et al.*. 2025. *EP241021a: A Months-duration X-Ray Transient with Luminous Optical and Radio Emission*. ApJL, 990, L29. arXiv:2505.07665.

81. Andrews, J. E., Shrestha, M., Bostroem, K. A., ..., **J. R. Farah**, *et al.*. 2025. *Asymmetries and Circumstellar Interaction in the Type II SN 2024bch*. ApJ, 980, 37. arXiv:2411.02497.

80. Li, W. X., Zhu, Z. P., Zou, X. Z., ..., **J. R. Farah**, *et al.*. 2025. *An extremely soft and weak*

fast X-ray transient associated with a luminous supernova. arXiv e-prints, arXiv:2504.17034.

79. Pastorello, A., Reguitti, A., Tartaglia, L., ..., **J. R. Farah**, *et al.*. 2025. *A long-lasting eruption heralds SN 2023ldh, a clone of SN 2009ip.* A&A, 701, A32. arXiv:2503.23123.

78. Baer-Way, R., Chandra, P., Modjaz, M., ..., **J. R. Farah**, *et al.*. 2025. *A Multiwavelength Autopsy of the Interacting Type IIn Supernova 2020ywx: Tracing Its Progenitor Mass-loss History for 100 Yr Before Death.* ApJ, 983, 101. arXiv:2412.06914.

77. Kumar, H., Berger, E., Blanchard, P. K., ..., **J. R. Farah**, *et al.*. 2025. *A Detection of Helium in the Bright Superluminous Supernova SN 2024rmj.* arXiv e-prints, arXiv:2506.06417.

76. Baczko, A. K., Kadler, M., Ros, E., ..., **J. R. Farah**, *et al.*. 2024. *The putative center in NGC 1052.* A&A, 692, A205. arXiv:2501.08685.

75. Pellegrino, C., Modjaz, M., Takei, Y., ..., **J. R. Farah**, *et al.*. 2024. *The X-Ray Luminous Type Ibn SN 2022ablq: Estimates of Preexplosion Mass Loss and Constraints on Precursor Emission.* ApJ, 977, 2. arXiv:2407.18291.

74. Dong, Y., Tsuna, D., Valenti, S., ..., **J. R. Farah**, *et al.*. 2024. *SN2023fyq: A Type Ibn Supernova with Long-standing Precursor Activity Due to Binary Interaction.* ApJ, 977, 254. arXiv:2405.04583.

73. Newsome, M., Arcavi, I., Howell, D. A., ..., **J. R. Farah**, *et al.*. 2024. *Mapping the Inner 0.1 pc of a Supermassive Black Hole Environment with the Tidal Disruption Event and Extreme Coronal-line Emitter AT 2022upj.* ApJ, 977, 258. arXiv:2406.11972.

72. Algaba, J., Baloković, M., Chandra, S., ..., **J. R. Farah**, *et al.*. 2024. *Broadband multi-wavelength properties of M87 during the 2018 EHT campaign including a very high energy flaring episode.* A&A, 692, A140. arXiv:2404.17623.

71. Szalai, T., Könyves-Tóth, R., Nagy, A., ..., **J. R. Farah**, *et al.*. 2024. *The story of SN 2021aatd: A peculiar 1987A-like supernova with an early-phase luminosity excess.* A&A, 690, A17. arXiv:2406.02498.

70. Dong, Y., Valenti, S., Ashall, C., ..., **J. R. Farah**, *et al.*. 2024. *Characterizing the Rapid Hydrogen Disappearance in SN 2022crv: Evidence of a Continuum between Type Ib and IIb Supernova Properties.* ApJ, 974, 316. arXiv:2309.09433.

69. Kumar, H., Berger, E., Hiramatsu, D., ..., **J. R. Farah**, *et al.*. 2024. *AT2023vto: An Exceptionally Luminous Helium Tidal Disruption Event from a Massive Star.* ApJL, 974, L36. arXiv:2408.01482.

68. Johnson, M. D., Akiyama, K., Baturin, R., ..., **J. R. Farah**, *et al.*. 2024. *The Black Hole Explorer: motivation and vision.* 13092, 130922D. arXiv:2406.12917.

67. Jacobson-Galán, W., Dessart, L., Davis, K., *et al.*. 2024. *Final Moments. II. Observational Properties and Physical Modeling of Circumstellar-material-interacting Type II Supernovae.* ApJ, 970, 189. arXiv:2403.02382.

66. Park, S., Rho, J., Yoon, S. C., ..., **J. R. Farah**, *et al.*. 2024. *Optical and Infrared Observation of a Type IIP Supernova 2021qqu.* 536, 165.

65. Faris, S., Arcavi, I., Makrygianni, L., ..., **J. R. Farah**, *et al.*. 2024. *Light-curve Struc-*

ture and Halpha Line Formation in the Tidal Disruption Event AT 2019azh. ApJ, 969, 104. arXiv:2312.03842.

64. Kwok, L. A., Siebert, M. R., Johansson, J., ..., **J. R. Farah**, *et al.*. 2024. *Ground-based and JWST Observations of SN 2022pul. II. Evidence from Nebular Spectroscopy for a Violent Merger in a Peculiar Type Ia Supernova.* ApJ, 966, 135. arXiv:2308.12450.

63. Event Horizon Telescope Collaboration, ..., **J. R. Farah**, *et al.*. 2024. *The persistent shadow of the supermassive black hole of M 87. I. Observations, calibration, imaging, and analysis.* A&A, 681, A79.

62. Pearson, J., Sand, D. J., Lundqvist, P., ..., **J. R. Farah**, *et al.*. 2024. *Strong Carbon Features and a Red Early Color in the Underluminous Type Ia SN 2022xkq.* ApJ, 960, 29. arXiv:2309.10054.

61. Siebert, M. R., Kwok, L. A., Johansson, J., ..., **J. R. Farah**, *et al.*. 2024. *Ground-based and JWST Observations of SN 2022pul. I. Unusual Signatures of Carbon, Oxygen, and Circumstellar Interaction in a Peculiar Type Ia Supernova.* ApJ, 960, 88. arXiv:2308.12449.

60. Yadavalli, S. K., Villar, V. A., Izzo, L., ..., **J. R. Farah**, *et al.*. 2024. *SN 2022oqm: A Bright and Multipeaked Calcium-rich Transient.* ApJ, 972, 194. arXiv:2308.12991.

59. Andrews, J. E., Pearson, J., Hosseinzadeh, G., ..., **J. R. Farah**, *et al.*. 2024. *SN 2022jox: An Extraordinarily Ordinary Type II SN with Flash Spectroscopy.* ApJ, 965, 85. arXiv:2310.16092.

58. Padilla Gonzalez, E., Howell, D., Terreran, G., ..., **J. R. Farah**, *et al.*. 2024. *SN 2022joj: A Potential Double Detonation with a Thin Helium Shell.* ApJ, 964, 196. arXiv:2308.06334.

57. Newsome, M., Arcavi, I., Howell, D. A., ..., **J. R. Farah**, *et al.*. 2024. *Probing the Subparsec Dust of a Supermassive Black Hole with the Tidal Disruption Event AT 2020mot.* ApJ, 961, 239. arXiv:2305.03767.

56. Paraschos, G., Kim, J. Y., Wielgus, M., ..., **J. R. Farah**, *et al.*. 2024. *Ordered magnetic fields around the 3C 84 central black hole.* A&A, 682, L3. arXiv:2402.00927.

55. Hiramatsu, D., Matsumoto, T., Berger, E., ..., **J. R. Farah**, *et al.*. 2024. *Multiple Peaks and a Long Precursor in the Type IIn Supernova 2021qqp: An Energetic Explosion in a Complex Circumstellar Environment.* ApJ, 964, 181. arXiv:2305.11168.

54. Raymond, A. W., Doeleman, S. S., Asada, K., ..., **J. R. Farah**, *et al.*. 2024. *First Very Long Baseline Interferometry Detections at 870 mum.* AJ, 168, 130. arXiv:2410.07453.

53. Event Horizon Telescope Collaboration, ..., **J. R. Farah**, *et al.*. 2024. *First Sagittarius A* Event Horizon Telescope Results. VIII. Physical Interpretation of the Polarized Ring.* ApJL, 964, L26.

52. Event Horizon Telescope Collaboration, ..., **J. R. Farah**, *et al.*. 2024. *First Sagittarius A* Event Horizon Telescope Results. VII. Polarization of the Ring.* ApJL, 964, L25.

51. Shrestha, M., Bostroem, K. A., Sand, D. J., ..., **J. R. Farah**, *et al.*. 2024. *Extended Shock Breakout and Early Circumstellar Interaction in SN 2024ggi.* ApJL, 972, L15. arXiv:2405.18490.

50. Shrestha, M., Pearson, J., Wyatt, S., ..., **J. R. Farah**, *et al.*. 2024. *Evidence of Weak Circumstellar Medium Interaction in the Type II SN 2023axu.* ApJ, 961, 247. arXiv:2310.00162.

49. Bora, Z., Könyves-Tóth, R., Vinkó, J., ..., **J. R. Farah**, *et al.*. 2024. *Ejecta Masses in Type Ia Supernovae—Implications for the Progenitor and the Explosion Scenario.* PASP, 136, 094201. arXiv:2408.11928.

48. Torne, P., Liu, K., Eatough, R. P., ..., **J. R. Farah**, *et al.*. 2023. *A Search for Pulsars around Sgr A* in the First Event Horizon Telescope Data Set.* ApJ, 959, 14. arXiv:2308.15381.

47. Bostroem, K. A., Pearson, J., Shrestha, M., ..., **J. R. Farah**, *et al.*. 2023. *Early Spectroscopy and Dense Circumstellar Medium Interaction in SN 2023ixf.* ApJL, 956, L5. arXiv:2306.10119.

46. Hosseinzadeh, G., Sand, D. J., Sarbadhicary, S. K., ..., **J. R. Farah**, *et al.*. 2023. *The Early Light Curve of SN 2023bee: Constraining Type Ia Supernova Progenitors the Apian Way.* ApJL, 953, L15. arXiv:2305.03071.

45. Hosseinzadeh, G., **J. R. Farah**, Shrestha, M., *et al.*. 2023. *Shock Cooling and Possible Precursor Emission in the Early Light Curve of the Type II SN 2023ixf.* ApJL, 953, L16. arXiv:2306.06097.

44. Bostroem, K. A., Dessart, L., Hillier, D. J., ..., **J. R. Farah**, *et al.*. 2023. *SN 2022acko: The First Early Far-ultraviolet Spectra of a Type IIP Supernova.* ApJL, 953, L18. arXiv:2305.01654.

43. Ben-Ami, T., Arcavi, I., Newsome, M., ..., **J. R. Farah**, *et al.*. 2023. *The Type Ibn Supernova 2019kbj: Indications for Diversity in Type Ibn Supernova Progenitors.* ApJ, 946, 30. arXiv:2212.03407.

42. Shrestha, M., Sand, D. J., Alexander, K. D., ..., **J. R. Farah**, *et al.*. 2023. *Limit on Supernova Emission in the Brightest Gamma-Ray Burst, GRB 221009A*. ApJL, 946, L25. arXiv:2302.03829.

41. Jorstad, S., Wielgus, M., Lico, R., ..., **J. R. Farah**, *et al.*. 2023. *The Event Horizon Telescope Image of the Quasar NRAO 530.* ApJ, 943, 170. arXiv:2302.04622.

40. Roelofs, F., Johnson, M. D., Chael, A., ..., **J. R. Farah**, *et al.*. 2023. *Polarimetric Geometric Modeling for mm-VLBI Observations of Black Holes.* ApJL, 957, L21. arXiv:2311.10695.

39. Hiramatsu, D., Tsuna, D., Berger, E., ..., **J. R. Farah**, *et al.*. 2023. *From Discovery to the First Month of the Type II Supernova 2023ixf: High and Variable Mass Loss in the Final Year before Explosion.* ApJL, 955, L8. arXiv:2307.03165.

38. Event Horizon Telescope Collaboration, ..., **J. R. Farah**, *et al.*. 2023. *First M87 Event Horizon Telescope Results. IX. Detection of Near-horizon Circular Polarization.* ApJL, 957, L20. arXiv:2311.10976.

37. Prather, B. S., Dexter, J., Moscibrodzka, M., ..., **J. R. Farah**, *et al.*. 2023. *Comparison of Polarized Radiative Transfer Codes Used by the EHT Collaboration.* ApJ, 950, 35. arXiv:2303.12004.

36. Broderick, A. E., Pesce, D. W., Gold, R., ..., **J. R. Farah**, *et al.*. 2022. *The Photon Ring in M87*.* ApJ, 935, 61. arXiv:2208.09004.

35. Issaoun, S., Wielgus, M., Jorstad, S., ..., **J. R. Farah**, *et al.*. 2022. *Resolving the Inner Parsec of the Blazar J1924-2914 with the Event Horizon Telescope.* ApJ, 934, 145. arXiv:2208.01662.

34. [S]Event Horizon Telescope Collaboration, ..., **J. R. Farah**, *et al.*. 2022. *First Sagittarius A* Event Horizon Telescope Results. VI. Testing the Black Hole Metric.* ApJL, 930, L17.

33. [S]Event Horizon Telescope Collaboration, ..., **J. R. Farah**, *et al.*. 2022. *First Sagittarius A* Event Horizon Telescope Results. V. Testing Astrophysical Models of the Galactic Center Black Hole.* ApJL, 930, L16.

32. [S]Event Horizon Telescope Collaboration, ..., **J. R. Farah**, *et al.*. 2022. *First Sagittarius A* Event Horizon Telescope Results. IV. Variability, Morphology, and Black Hole Mass.* ApJL, 930, L15. arXiv:2311.08697.

31. [S]Event Horizon Telescope Collaboration, ..., **J. R. Farah**, *et al.*. 2022. *First Sagittarius A* Event Horizon Telescope Results. III. Imaging of the Galactic Center Supermassive Black Hole.* ApJL, 930, L14.

30. [S]Event Horizon Telescope Collaboration, ..., **J. R. Farah**, *et al.*. 2022. *First Sagittarius A* Event Horizon Telescope Results. II. EHT and Multiwavelength Observations, Data Processing, and Calibration.* ApJL, 930, L13.

29. [S]Event Horizon Telescope Collaboration, ..., **J. R. Farah**, *et al.*. 2022. *First Sagittarius A* Event Horizon Telescope Results. I. The Shadow of the Supermassive Black Hole in the Center of the Milky Way.* ApJL, 930, L12.

28. [S]Broderick, A. E., Gold, R., Georgiev, B., ..., **J. R. Farah**, *et al.*. 2022. *Characterizing and Mitigating Intraday Variability: Reconstructing Source Structure in Accreting Black Holes with mm-VLBI.* ApJL, 930, L21.

27. [S]Georgiev, B., Pesce, D. W., Broderick, A. E., ..., **J. R. Farah**, *et al.*. 2022. *A Universal Power-law Prescription for Variability from Synthetic Images of Black Hole Accretion Flows.* ApJL, 930, L20.

[S]This is the first Galactic Center results series from the Event Horizon Telescope, presenting the second image of a black hole and the first horizon-scale images of Sgr A*.

26. Satapathy, K., Psaltis, D., Özel, F., ..., **J. R. Farah**, *et al.*. 2022. *The Variability of the Black Hole Image in M87 at the Dynamical Timescale.* ApJ, 925, 13. arXiv:2111.01317.

25. Wielgus, M., Marchili, N., Martí-Vidal, I., ..., **J. R. Farah**, *et al.*. 2022. *Millimeter Light Curves of Sagittarius A* Observed during the 2017 Event Horizon Telescope Campaign.* ApJL, 930, L19. arXiv:2207.06829.

24. Wielgus, M., Moscibrodzka, M., Vos, J., ..., **J. R. Farah**, *et al.*. 2022. *Orbital motion near Sagittarius A* . Constraints from polarimetric ALMA observations.* A&A, 665, L6. arXiv:2209.09926.

23. Janssen, M., Falcke, H., Kadler, M., ..., **J. R. Farah**, *et al.*. 2021. *Event Horizon Telescope observations of the jet launching and collimation in Centaurus A.* Nature Astronomy, 5, 1017-1028. arXiv:2111.03356.

22. Narayan, R., Palumbo, D. C., Johnson, M. D., ..., **J. R. Farah**, *et al.*. 2021. *The Polarized Image of a Synchrotron-emitting Ring of Gas Orbiting a Black Hole.* ApJ, 912, 35. arXiv:2105.01804.

21. Kocherlakota, P., Rezzolla, L., Falcke, H., ..., **J. R. Farah**, *et al.*. 2021. *Constraints on black-hole charges with the 2017 EHT observations of M87*.* Phys. Rev. D, 103, 104047. arXiv:2105.09343.

20. Goddi, C., Martí-Vidal, I., Messias, H., ..., **J. R. Farah**, *et al.*. 2021. *Polarimetric Properties of Event Horizon Telescope Targets from ALMA.* ApJL, 910, L14. arXiv:2105.02272.

19. Event Horizon Telescope Collaboration, ..., **J. R. Farah**, *et al.*. 2021. *First M87 Event Horizon Telescope Results. VIII. Magnetic Field Structure near The Event Horizon.* ApJL, 910, L13. arXiv:2105.01173.

18. Event Horizon Telescope Collaboration, ..., **J. R. Farah**, *et al.*. 2021. *First M87 Event Horizon Telescope Results. VII. Polarization of the Ring.* ApJL, 910, L12. arXiv:2105.01169.

17. Raymond, A. W., Palumbo, D., Paine, S. N., ..., **J. R. Farah**, *et al.*. 2021. *Evaluation of New Submillimeter VLBI Sites for the Event Horizon Telescope.* ApJS, 253, 5. arXiv:2102.05482.

16. Group, E. M. S. W., Algaba, J., Anczarski, J., ..., **J. R. Farah**, *et al.*. 2021. *Broadband Multi-wavelength Properties of M87 during the 2017 Event Horizon Telescope Campaign.* ApJL, 911, L11. arXiv:2104.06855.

15. Kim, J. Y., Krichbaum, T. P., Broderick, A. E., *et al.*. 2020. *Event Horizon Telescope imaging of the archetypal blazar 3C 279 at an extreme 20 microarcsecond resolution.* A&A, 640, A69.

14. Johnson, M. D., Lupsasca, A., Strominger, A., ..., **J. R. Farah**, *et al.*. 2020. *Universal interferometric signatures of a black hole's photon ring.* Science Advances, 6, eaaz1310. arXiv:1907.04329.

13. Fabbiano, G., Elvis, M., Accomazzi, A., *et al.*. 2019. *Increasing the Discovery Space in Astrophysics: The Exploration Question for Stars and Stellar Evolution.* BAAS, 51, 85.

12. Fabbiano, G., Elvis, M., Accomazzi, A., *et al.*. 2019. *Increasing the Discovery Space in Astrophysics: The Exploration Question for Resolved Stellar Populations.* BAAS, 51, 84.

11. Fabbiano, G., Elvis, M., Accomazzi, A., *et al.*. 2019. *Increasing the Discovery Space in Astrophysics: The Exploration Question for Planetary Systems.* BAAS, 51, 86.

10. Fabbiano, G., Elvis, M., Accomazzi, A., *et al.*. 2019. *Increasing the Discovery Space in Astrophysics: The Exploration Question for Galaxy Evolution.* BAAS, 51, 87.

9. Fabbiano, G., Elvis, M., Accomazzi, A., *et al.*. 2019. *Increasing the Discovery Space in Astrophysics: The Exploration Question for Cosmology.* BAAS, 51, 88.

8. Fabbiano, G., Elvis, M., Accomazzi, A., *et al.*. 2019. *Increasing the Discovery Space in Astrophysics: The Exploration Question for Compact Objects.* BAAS, 51, 89.

7. Fabbiano, G., Elvis, M., Accomazzi, A., *et al.*. 2019. *Increasing the Discovery Space in Astrophysics - A Collation of Six Submitted White Papers.* arXiv e-prints, arXiv:1903.06634.

6. †Event Horizon Telescope Collaboration, ..., **J. R. Farah**, *et al.*. 2019. *First M87 Event Horizon Telescope Results. VI. The Shadow and Mass of the Central Black Hole.* ApJL, 875, L6. arXiv:1906.11243.

5. †Event Horizon Telescope Collaboration, ..., **J. R. Farah**, *et al.*. 2019. *First M87 Event Horizon Telescope Results. V. Physical Origin of the Asymmetric Ring.* ApJL, 875, L5. arXiv:1906.11242.

4. †Event Horizon Telescope Collaboration, ..., **J. R. Farah**, *et al.*. 2019. *First M87 Event Horizon Telescope Results. IV. Imaging the Central Supermassive Black Hole.* ApJL, 875, L4. arXiv:1906.11241.

3. †Event Horizon Telescope Collaboration, ..., **J. R. Farah**, *et al.*. 2019. *First M87 Event Horizon Telescope Results. III. Data Processing and Calibration.* ApJL, 875, L3. arXiv:1906.11240.

2. †Event Horizon Telescope Collaboration, ..., **J. R. Farah**, *et al.*. 2019. *First M87 Event Horizon Telescope Results. II. Array and Instrumentation.* ApJL, 875, L2. arXiv:1906.11239.

1. †Event Horizon Telescope Collaboration, ..., **J. R. Farah**, *et al.*. 2019. *First M87 Event Horizon Telescope Results. I. The Shadow of the Supermassive Black Hole.* ApJL, 875, L1. arXiv:1906.11238.

†These six papers are the first results papers from the Event Horizon Telescope, presenting the first image of a black hole.

Conference publications

(last updated: Aug 2025)

4. M. D. Johnson, K. Akiyama, R. Baturin, ... **J. Farah** et al. (2024). *The Black Hole Explorer: motivation and vision.* In: Space Telescopes and Instrumentation 2024: Optical, Infrared, and Millimeter Wave (SPIE Conf. Ser. 13092), 130922D.

3. Haworth, K., Johnson, M. D., Pesce, D. W., ... **J. R. Farah** et al. 2019. *Studying black holes on horizon scales with space-VLBI.* arXiv:1909.01405

2. Blackburn, L., Doeleman, S., Dexter, J., ... **J. R. Farah** et al. 2019. *Studying Black Holes on Horizon Scales with VLBI Ground Arrays.* arXiv:1909.01411

1. Fabbiano, E., Berriman, B., Bose, C., ... **J. R. Farah**, et al. 2019. *Increasing the Discovery Space in Astrophysics - A Collation of Six Submitted White Papers.*

Technical publications

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5. Chael A., Bouman, K., Johnson M., Wielgus M., Blackburn L., Chan C., **Farah J.**, Palumbo D., Pesce D., et al. 2022. *eht-imaging: v1.2.4: Imaging interferometric data with regularized maximum likelihood.* Zenodo (software) publication.

4. Chael A., Bouman, K., Johnson M., Wielgus M., Blackburn L., Chan C., **Farah J.**, Palumbo D., Pesce D. 2019. *eht-imaging: v1.1.0: Imaging interferometric data with regularized maximum likelihood.* Zenodo (software) publication.

3. **Farah J**, Felt N, Franklin M, Giorimini P, Rogan C, Tuna A, Wang A. 2017. *All The Sparks We Cannot See.* ATLAS Internal Note.

2. **Farah J**, Felt N, Frank, Giorimini P, Rogan C, Tuna A, Wang A. 2017. *Test of the Micromegas Trigger Processor with Cosmic Ray Muons.* ATLAS Internal Note.

1. Felt N, Franklin M, Giorimini P, Rogan C, Tuna A, Wang A, **Farah J**. 2017. *Performance of a Micromegas octuplet after removing the major cause of MMFE8 noise.* ATLAS Internal Note.

Classifications, telegrams, and circulars

(last updated: Aug 2025)

First or co-author on 90+ Transient Name Server classification reports, AstroNotes, and GCN circulars, 2022–present, as part of the Global Supernova Project.